EXPERIMENT-13

Greatest of 2 numbers.

AIM: To write an Assembly language Program to find the smallest number in an array using 8085 Microprocessor in GNUsim.

Software used: GNUsim8085

Algorithm:

1. Initialize the count.
2. Get the input numbers
3. Compare the content of Accumulator (A) with HL pair for all input numbers
4. Stores the smallest number in the output register.
5. ~~Eng~~ End the program.

Program:

```
LDA 2050 - load memory to A.
MOV B,A - copy A to B
LDA 2051 - load memory to A.
CMP B - Compare A with B.
JNC STORE - Jump if no carry
MOV A,B - copy B to A.
STA 2052 - Store result in memory.
HLT - Stop program.
```

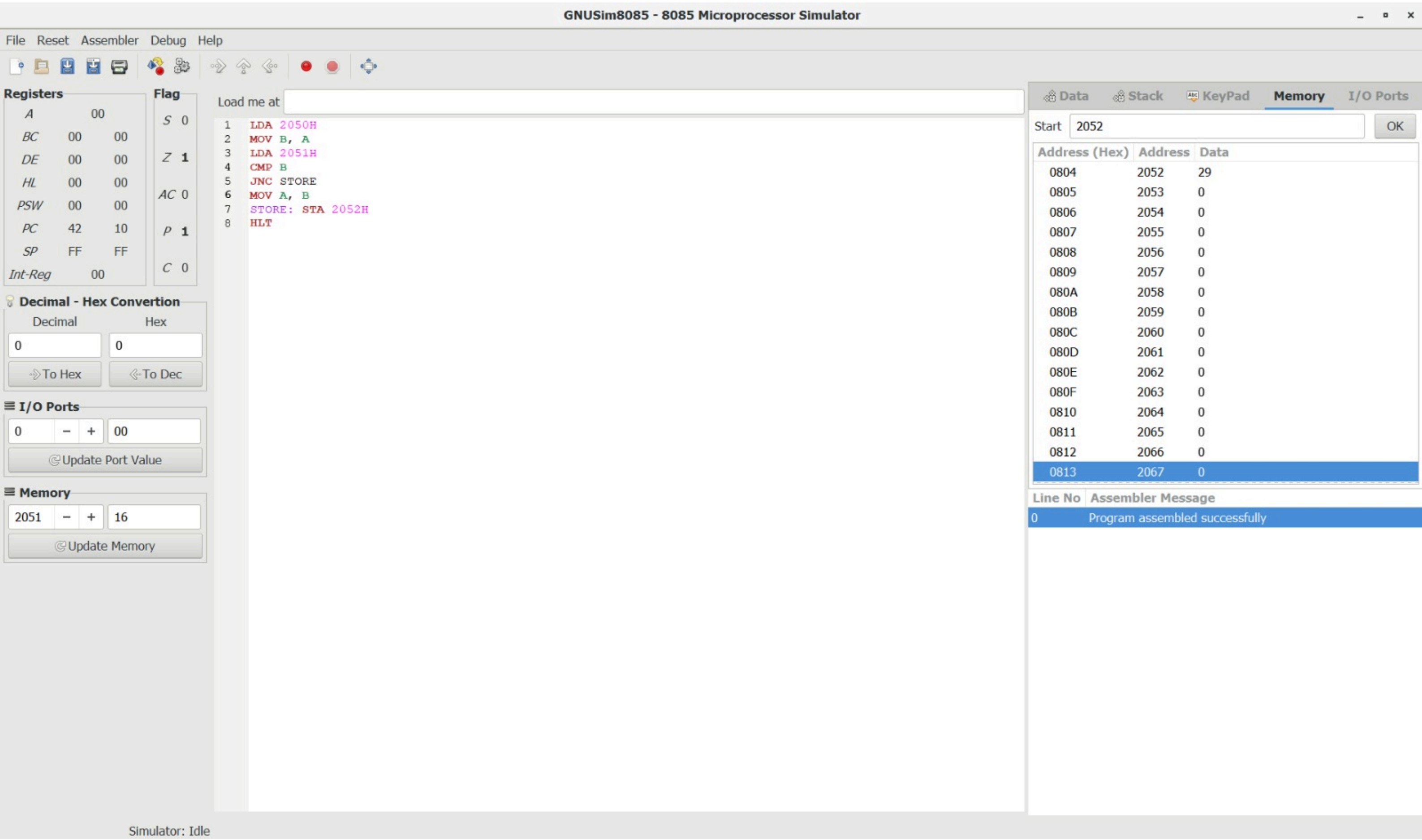
Input:

Address	data
2050	29
2051	22

Output:

Address	data
2052	29

Result: Thus the Assembly language Program to find the smallest number in an array using 8085 Microprocessor in GNUsim is performed.


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 00
BC 00 00
DE 00 00
HL 00 00
PSW 00 00
PC 42 10
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Load me at
LDA 2050H
MOV B, A
LDA 2051H
CMP B
JNC STORE
MOV A, B
STORE: STA 2052H
HLT
Decimal - Hex Convertion
Decimal
Hex
To Hex
To Dec
I/O Ports
Update Port Value
Memory
2051
16
Update Memory
Data Stack KeyPad Memory I/O Ports
Start 2052
OK
Address (Hex) Address Data
0804 2052 29
0805 2053 0
0806 2054 0
0807 2055 0
0808 2056 0
0809 2057 0
080A 2058 0
080B 2059 0
080C 2060 0
080D 2061 0
080E 2062 0
080F 2063 0
0810 2064 0
0811 2065 0
0812 2066 0
0813 2067 0
Line No Assembler Message
0 Program assembled successfully
Simulator: Idle